

Jorge Castano

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EDUCATION

CARNEGIE MELLON UNIVERSITY, Pittsburgh PA

May 8th, 2026

Bachelor of Science in Mechanical Engineering

PROJECTS

3D PRINTED 3 DOF ROBOTIC ARM WITH GRIPPER | Project Lead

May 2026

- Designed a 3-DOF robotic arm (base, shoulder, elbow, gripper) in Onshape with integrated mounting for micro-servos and Arduino Uno, optimizing geometry for wire routing, modular assembly, and mass reduction.
- Implemented inverse kinematics (IK) to compute joint angles from end-effector coordinates, enabling coordinated 2D positioning and programmed a finite state machine (FSM) for deterministic pick-and-place sequences.
- Applied PID control for smooth servo motion and integrated multiple control interfaces (potentiometer teleoperation, serial commands) for initial hardware-software integration, real-time response, and smooth servo acceleration and positioning accuracy.

AUTOMATED BLOOD BAG SEALER | Project Lead

March 2026

- Applied a formal engineering design process, utilizing House of Quality and Design Matrices to translate customer needs into verifiable engineering metrics and down-select the final design concept.
- Designed a dual-axis positioning gantry featuring manual horizontal adjustment for different blood bag sizes and an actively controlled vertical drive using a wide gear and rack to precisely position the sealing mechanism over the tube within a 2 mm tolerance.
- Integrated a limit switch to trigger an automated sealing cycle and developed a pinching jaw assembly that uses nichrome wire to simultaneously pinch and heat the tube, creating a sterile seal.
- Delivered a functional prototype demonstrating a repeatable, automated sealing process, reducing user interaction and potential variability compared to manual methods.

MECHANICAL SYSTEM DYNAMIC PARAMETER CHARACTERIZATION | Lab Lead

February 2026

- Performed controlled experiments on oscillating mechanical systems to capture raw displacement and force encoder data during free and forced vibration that was processed in MATLAB.
- Developed scripts to plot experimental data, compute the logarithmic decrement, and extract key dynamic properties (damping ratio, natural frequency, spring stiffness) from time-response data.
- Implemented open and closed loop controllers for rectilinear and torsional two degree of freedom systems and tuned proportional, integral, and derivative gain in order to reduce experimental overshoot, rise time, peak time, and steady state error.

SMART BENCH PRESS FORM ANALYZER | Computer Vision Lead & Testing Coordinator

February 2025

- Collaborated with a multidisciplinary team to design and implement a real-time bench press form monitoring system, aimed at ensuring proper flat bench press lifting technique and reducing risk of injury.
- Implemented OpenCV in Python to analyze barbell position during flat bench press, determining the angle of the barbell and whether or not the barbell remained within a 5 degree tolerance range parallel to the ground.
- Integrated electromyography (EMG) sensors to collect muscle activation data during proper repetitions, creating a reference dataset for future analysis and live personalized feedback for the user through use of a speaker.
- Processed raw EMG signals using Arduino by implementing real-time filtering, rectification, and smoothing algorithms to extract meaningful muscle activation data during bench press repetitions.

WORK EXPERIENCE

DEADWOOD CUSTOM | Solidworks Technical Drafter & CNC Operator

February 2025 - June 2025

- Led design and development of detailed 3D models, technical drawings, and fabrication schematics for custom retail displays and fixtures using SolidWorks, ensuring adherence to client branding and functional requirements.
- Operated and programmed a CNC router using VCarve Pro to cut DXF 2D projections of 3D models to ensure dimensional accuracy.
- Optimized design for ready-to-use and customer-assembled-configurations through modular components to fit shipping containers or other space-constrained logistics - reducing freight cost and simplifying on-site installation.

TECHNICAL SKILLS

Engineering Software: ANSYS Workbench, ANSYS Mechanical/Thermal/Fluent, SolidWorks, MATLAB, Simulink, Onshape
Programming: Python, OpenCV, PyTorch, C, C++